

Exercise

Chapter 2: Statistical learning

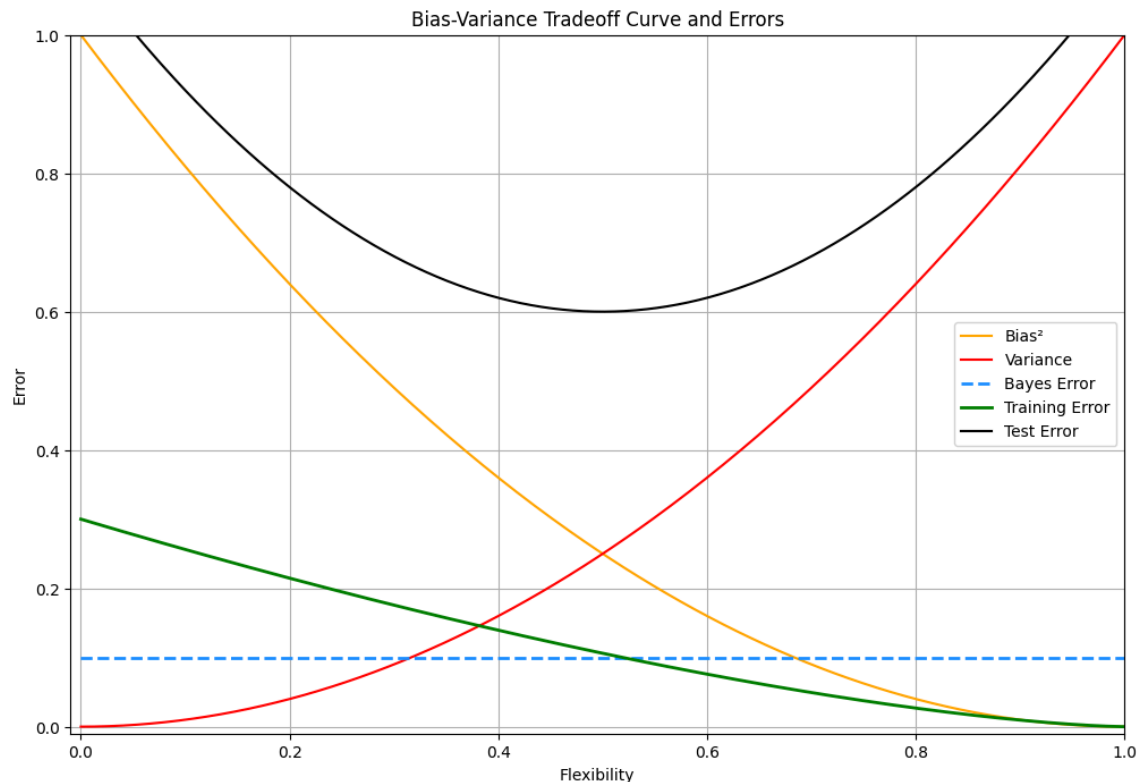
Question 1.

- a) Flexible methods would be better when the sample size n is extremely large than the predictors p . There is plenty of data points to fit without overfitting.
- b) Flexible methods would be worse when p is large than the number of observations n . This will make the flexible method to overfit, hence inflexible methods is preferred.
- c) Non-linear relationships typically require flexible statistical methods in which it would be better to fit in every observation.
- d) Inflexible methods would be better, simpler models ignore this irreducible error and focus on the main trends.

Question 2.

- a) This is a regression problem interested in inference because we're need to understanding the CEO salary. n is the top 500 firms and p is the records that is profit, no of employees etc.
- b) Classification problem with interest in prediction, We care about accuracy of the future outcome. n is the 20 similar products, p is the price, marketing budget and ten other variables.
- c) Regression problem and predicting the outcome. Based on the data collected, the how does not concern us. n is the weekly data collected in 2012 and p is the currency change in those countries stated.

Question 3



Explaining the Curve:

Bias: Error introduced by approximating a real-life problem. So increasing flexibility may reduce bias till a certain point where there will be no more changes.

Variance: Refers to how much change there would be if we used a different datasets. High flexible models where the model fits on every observation causes high variance.

Training Error: The rate of mistakes the model makes on the exact dataset used to train. Flexible methods have low error rates.

Test Error: The rate of mistake the model make on a brand new or unseen dataset.

Bayes Error: The *Minimum possible error* that is cause by noise in the data, no model can go beyond this. It is a straight line because it does not depend on the model flexibility.